

PHAM ANH TUAN

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EDUCATION

National University of Singapore | *GPA: 4.82/5.0* *Aug 2023 - present*
Singapore
B. Eng. | *Robotics and Machine Intelligence (2nd Major: Quantitative Finance, Minor: AI)*
Relevant Coursework: Calculus, Differential Equations, Linear Algebra, Fundamentals of Robotics, Design and Manufacturing, Fundamentals of Artificial Neural Network, Programming Methodology
Ly Tu Trong High School for the Gifted | *GPA: 9.7/10.0* *August 2022 – May 2025*
Can Tho, Vietnam
Leadership: Class President; Founder & President, Chess Club; Vice President, Mathematics Club

HONORS & AWARDS

- Silver Medal**, Singapore Inter-Varsity Chess League, SIVCA *February 2026*
- 4th place**, RoboRoarZ AI Challenge, Singapore University of Technology and Design *January 2026*
- 3 Bronze Medals**, Vietnamese Championship for Chess Clubs, Vietnam Chess Federation *June 2025*
- Bronze Medal**, Vietnam National Physics Olympiad (VPhO), Ministry of Education *January 2025*
- 3rd Prize**, City Contest for Youth & Children Innovation *September 2024*

EXPERIENCE

Software Engineer *September 2025 - present*
Singapore
Bumblebee Autonomous Systems, National University of Singapore

- Hornet 11.0 AUV project:**
 - Improved object recognition accuracy by 22%** via a **novel automated dataset curation pipeline**.
 - Implemented a **real-time stereo camera depth pipeline**, achieving **≤2% error** through rigorous calibration & validation.
 - Built a robust **depth estimation system** using **MAD-based background filtering & symmetry correction** to localize 3D objects from partial disparity maps.
 - Reduced ROS2 communication overhead** by designing **custom message types** with **minimal bounding box & class metadata**.
 - Designed a competition-ready task system using **Py_trees** to build **behavior trees** that handled complex robot decision making.

SELECTED PROJECTS

Image Inpainting *April 2026*

- Designed & trained **CAE** and **Vision Transformer** architectures for **image inpainting** on **CIFAR-10** & **Oxford-IIIT Pet**, benchmarking optimizer, latent dimensionality, and masking strategies.
- Engineered a **7-phase curriculum training schedule** with **adaptive MAE + perceptual loss weighting**, enabling coherent reconstructions under **69% random occlusion**.
- Validated **cross-domain generalisation** to unseen object categories, confirming the model learned general inpainting priors rather than texture memorisation.

Hospitality AI *January 2026*

- Implemented a **low-latency, multi-stage ML pipeline** using **ResNet-18** & **YOLOv8n** for real-time bed quality inspection.
- Built **augmentation methods** to handle real-world constraints (occlusions, lighting, partial views), **improving robustness by 67%**.
- Resolved training plateaus** through targeted **hyperparameter tuning & learning-rate scheduling**.

Realistic AI Assistant *December 2025*

- Built a **conversational robot** on **ESP32-S3** using **WebSocket** audio streaming, **I2S microphone & amplifier** for **real-time STT/TTS**.
- Engineered **3D-printed animatronic eyes** controlled by a **PCA9685 16-channel PWM driver** & **custom LCD lip-sync patterns**.
- Coordinated **ESP32-Arduino state management** through **GPIO bit-signaling** for reliable system control.

Wall-Following Robot *November 2025*

- Implemented **autonomous wall-following & maze escape** in **ROS2** using **Heading & Pledge algorithms** with **RANSAC-filtered LiDAR** for robust detection.
- Optimized **holonomic drive kinematics** in **Gazebo**, achieving the **fastest course time**, deployed in simulation & real hardware.

SKILLS

Technical experience & interests: Robotics Perception & Planning, Autonomous Systems, Robotic Simulation & Control
Programming languages: Python, C/C++, HTML/CSS
Engineering tools: ROS, RViz, Gazebo, Linux, Git, MATLAB, LaTeX, Docker, SSH, Foxglove, PyTorch, YOLO, OpenCV, Py_trees
Languages: Vietnamese (Native), English (Fluent — IELTS 8.5)
Certifications & Activities: SAT: 1540/1600, AP Physics 1 & 2: 5/5, 1st Dan Taekwondo, Soccer, Piano & Drums, Chess